

Q1:

Date _____ 20 _____

SR	x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
1	0	10	-50	-30	1500	2500	900
2	25	25	-25	-15	375	625	225
3	50	40	0	0	0	0	0
4	75	55	25	15	375	625	225
5	100	70	50	30	1500	2500	900
Sum	250	200			3750	6250	2250
Mean	50	40					

$$y = mx + b$$

$$m = r \cdot \frac{\sigma_y}{\sigma_x} \Rightarrow \sigma_y = \sqrt{\frac{\sum (y - \bar{y})^2}{N-1}} = \sqrt{\frac{2250}{4}} = \boxed{23.71}$$

$$\sigma_x = \sqrt{\frac{\sum (x - \bar{x})^2}{N-1}} = \sqrt{\frac{6250}{4}} = \boxed{39.52}$$

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}} = \frac{3750}{\sqrt{6250 \times 2250}} = \frac{3750}{3750} = \boxed{1}$$

$$m = r \cdot \frac{\sigma_y}{\sigma_x} \Rightarrow m = (1) \cdot \frac{23.71}{39.52} = \boxed{0.599}$$

$$b = \text{mean}(y) - m * (\text{mean}(x))$$

$$40 - (0.599)(50)$$

$$b = 40 - 29.95 = \boxed{10.05}$$

$$y = mx + b, \text{ if } x = 125$$

then $y = (0.599)(125) + 10.05$

$$= 74.875 + 10.05$$

$$= 84.925$$

$$y = \boxed{\approx 85}$$

	x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
1	15	10	-15	-10	150	225	100
2	20	25	-10	-5	50	100	25
3	35	20	+5	0	0	25	0
4	50	35	+20	15	300	400	225
Sum	120	80			500	750	350
Mean	30	20					

$$y = mx + b$$

$$m = r \cdot \frac{\sigma_y}{\sigma_x}, \quad \sigma_y = \sqrt{\frac{\sum (y - \bar{y})^2}{N-1}} = \sqrt{\frac{350}{3}}$$

$$\sigma_y = \sqrt{116.66} = 10.800$$

$$\sigma_x = \sqrt{\frac{\sum (x - \bar{x})^2}{N-1}} = \sqrt{\frac{750}{3}} = \sqrt{250} = 15.811$$

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2} \sqrt{\sum (y - \bar{y})^2}} = \frac{500}{\sqrt{750 \times 350}} = \frac{500}{512.34} = 0.97$$

$$m = 0.97 \left(\frac{10.800}{15.811} \right) = \frac{10.5398}{15.811} = 0.66$$

$$b = \text{mean}(y) - m * (\text{mean}(x))$$

$$20 - (0.66)(30)$$

$$20 - 19.8 = 0.2$$

$$y = mx + b, \text{ if } x = 55.$$

$$y = (0.66)(55) + 0.2$$

$$= 36.5$$